



Skill-Based Assignment: Data Analysis and Visualization with Python

Objective:

In this skill-based assignment (SBA), you are expected to analyze and visualize the given dataset using Python, Pandas, Matplotlib, and other supporting libraries.

Time/Duration:

- You have **3 hours** to complete this SBA.
- Please ensure that you are not interrupted during the test.

Passing Criteria

- **Total Points: 100.**
- **Passing Score: 70% (70 points).**

Instructions

Section One

Use the following CSV file for this SBA:

→ [Click to download Sample Data. "woocommerce-product-export.csv"](#)

Below are the Tasks:

1. Read Data from the [woocommerce-product-export.csv](#) file using the python library.
2. Show a concise summary of the columns using the info() method.
3. Show a summary of statistics pertaining to the columns.
4. Print the first five rows by default.
5. Print the last five rows by default.
6. Print the **"total_profit"** and **"month_number"** columns only.
7. **Read the total profit of all months and show it using the Bar plot.**

➤ The generated **Bar plot** must include the following properties:

- ❖ X label name = Month number.
- ❖ Y label name = Total profit.
- ❖ Title = Company profit per month.

The graph should look like the screenshot below:



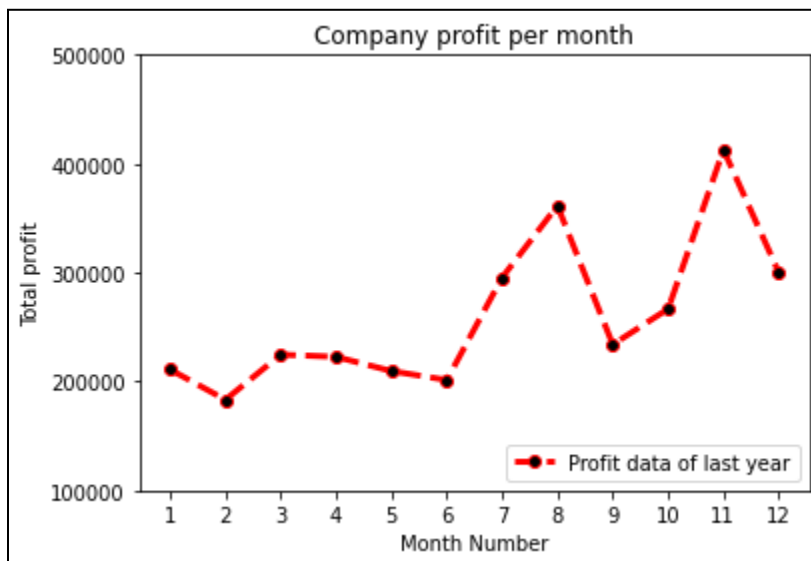
8. **Read the total profit of all of the months, and show the line plot with the following style properties:**

➤ The generated **line plot** must include the following Style properties:



- ❖ The line style should be **dotted** and the **line color** should be **red**.
- ❖ Show the legend at the lower-right location.
- ❖ X label name = Month number
- ❖ Y label name = Total profit.
- ❖ Title = Company profit per month.
- ❖ Add a circle marker; the line marker color = black.
- ❖ The line width should be 3.

The graph should look like the screenshot below:



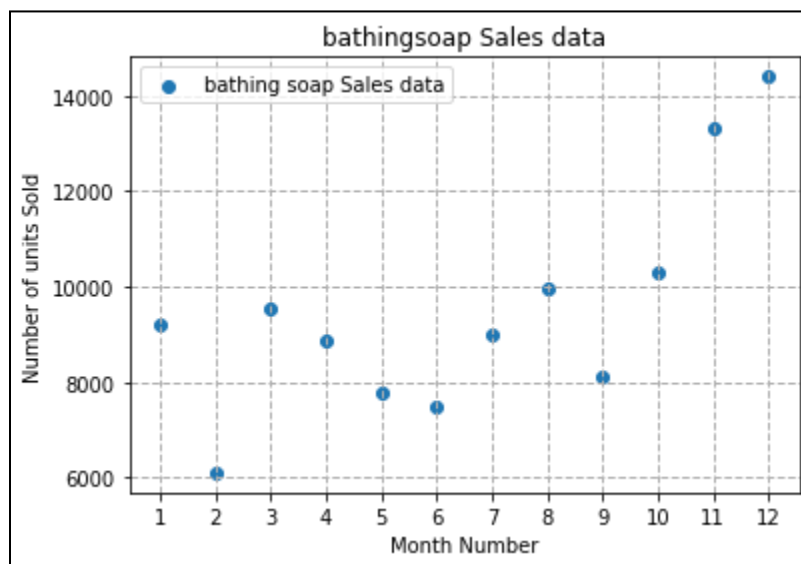
9. Print all of the product sales data and show it using a multi-line plot.

- Display the number of units sold per month for each product using multi-line plots. (i.e., show a separate plotline for each product).
- The graph should look like the screenshot below:



10. Read “*bathingsoap*” sales data for each month and show it using a scatter plot.

- Add a grid to the plot. The grid line style should be “-”.
- The scatter plot should look like the screenshot below:





Section Two

Below are the tasks:

1. **Given Dataset:**

date=["25/12","26/12","27/12"]

temp=[8.5,10.5,6.8]

A **line** chart of date versus temperature must include the following properties:

- X label name = Date
- Y label name = temperature
- Title = Date-wise Temperature

2. **Given Dataset:**

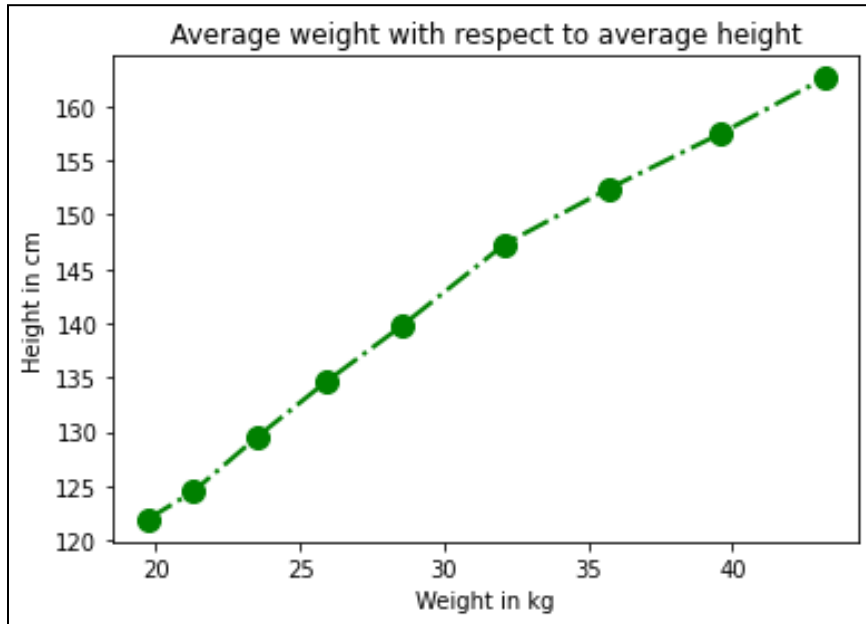
height=[121.9,124.5,129.5,134.6,139.7,147.3,152.4,157.5,162.6]

weight=[19.7,21.3,23.5,25.9,28.5,32.1,35.7,39.6,43.2]

Show the average weight against the average height (use a line chart or any other chart):

- The generated **line plot** must include the following Style properties:
 - ❖ The line style should be dash-dot and the line color should be green.
 - ❖ Show the legend at the lower-right location.
 - ❖ X label name = Weight in kg.
 - ❖ Y label name = Height in cm.
 - ❖ Title = Average weight with respect to average height.
 - ❖ Add a circle marker: marker size = 10; circle marker color = green.

The graph should look like the screenshot below:



Grading Summary

Category	Points
Section One	65
Section Two	25
Project Submission	10
Total	100 Points

Scoring Summary

Level	Score Range	Descriptor
Excellent	90–100	All plots and code steps are accurate, well-documented, and fully meet formatting/style requirements.
Proficient	80–89	Most outputs are accurate with minor formatting or style errors. Documentation is clear.
Satisfactory	70–79	Meets minimum technical criteria; minor omissions or visual inaccuracies present.
Fail	Below 70	Missing required plots, incorrect outputs, poor documentation, or incomplete submission.

SBA submission guidelines

- Submit either an **.ipynb (Jupyter Notebook)** file or a **.py (Python script)** file.
 - The file must be named: **YOURNAME_386_SBA**.
 - Ensure that each step is **properly documented** with comments.
- Create a Word document.
 - Name the word document: **YOURNAME_386_DOC_SBA.doc**.
 - Take a screenshot of every result or chart.
 - Include all screenshots in a Word document.
 - Put any necessary information and details in the document, if needed.
 - Upload the Word document to Canvas.